1. Differentiability implies continuity

Theorem
If a function f is differentiable at a then f is continuous at a.

(So the theorem says if $\lim_{x \to a} \frac{f(x)-f(a)}{x-a} = f'(a)$ (a finite number) then $\lim_{x \to a} f(x) = f(a)$)

Proof
$\lim_{x \to a} f(x)$
$= \lim_{x \to a} (f(x) - f(a) + f(a))$ Addition or subtraction of the same term
$= \lim_{x \to a} (f(x) - f(a)) + \lim_{x \to a} f(a)$ Limit law of addition
$= \lim_{x \to a} \frac{f(x)-f(a)}{x-a} (x-a) + f(a)$ Multiplication and division by the same term
$= \lim_{x \to a} \frac{f(x)-f(a)}{x-a} \cdot \lim_{x \to a} (x-a) + f(a)$ Limit law of multiplication
$= f'(a) \cdot 0 + f(a)$ f is differentiable
$= f(a)$

So $\lim_{x \to a} f(x) = f(a)$ and therefore f is continuous in $x = a$.

Note: If f is continuous in a then f is not necessarily differentiable in a.
Counter example: $f(x) = |x|$ is continuous in 0 but not differentiable in 0.

2. (i) A positive derivative indicates that the function is increasing.

Definition of an increasing function:
A function f is increasing on an interval I if $f(x_2) > f(x_1)$ for any $x_1, x_2 \in I$ such that $x_2 > x_1$.

Theorem: If $f' > 0$ on an interval I then f is increasing on the interval.

Proof: Pick any two points $x_1, x_2 \in I$ such that $x_2 > x_1$.
Consider the function f on the interval $[x_1, x_2]$.

1. f is continuous on $[x_1, x_2]$ (f is differentiable on I)
2. f is differentiable on (x_1, x_2) (f is differentiable on I)

So, according to the Mean Value Theorem a $c \in (x_1, x_2)$ exists such that

$$f'(c) = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

But $f'(c) > 0$ and $x_2 - x_1 > 0$, therefore $f(x_2) - f(x_1) > 0$.
It follows that $f(x_2) > f(x_1)$ and therefore f is increasing on I.

2 (ii) A negative derivative indicates that the function is decreasing.

Definition of an decreasing function:
A function f is decreasing on an interval I if $f(x_2) < f(x_1)$ for any $x_1, x_2 \in I$ such that $x_2 > x_1$.

Theorem: If $f' < 0$ on an interval I then f is decreasing on the interval.

The proof of this theorem is similar to that given above.

3. Proof of the product rule for differentiation

$$\frac{d}{dx}(f(x).g(x)) = \lim_{h\to 0} \frac{f(x+h)g(x+h)-f(x)g(x)}{h}$$

$$= \lim_{h\to 0} \frac{f(x+h)g(x+h)-f(x)g(x+h)+f(x)g(x+h)-f(x)g(x)}{h}$$ (Subtracting and adding the term $f(x)g(x+h)$)

$$= \lim_{h\to 0} \frac{(f(x+h)-f(x))g(x+h)+f(x)(g(x+h)-g(x))}{h}$$

$$= \lim_{h\to 0} \frac{(f(x+h)-f(x))g(x+h)}{h} + \lim_{h\to 0} \frac{f(x)(g(x+h)-g(x))}{h}$$

$$= \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} \lim_{h\to 0} g(x+h) + \lim_{h\to 0} f(x) \lim_{h\to 0} \frac{g(x+h)-g(x)}{h}$$

$$= f'(x)g(x) + f(x)g'(x)$$

4. Derivative of $y = \sin x$

$$\frac{d}{dx}\sin x = \lim_{h\to 0} \frac{\sin(x+h)-\sin x}{h}$$

$$= \lim_{h\to 0} \frac{\sin x\cos h+\cos x\sin h-\sin x}{h}$$

$$= \lim_{h\to 0} \frac{\sin x(\cos h-1)+\cos x\sin h}{h}$$

$$= \sin x \lim_{h\to 0} \frac{(\cos h-1)}{h} + \cos x \lim_{h\to 0} \frac{\sin h}{h}$$

$= \cos x$ because $\lim_{h\to 0} \frac{(\cos h-1)}{h} = 0$ and $\lim_{h\to 0} \frac{\sin h}{h} = 1$

We show why $\lim_{h\to 0} \frac{(\cos h-1)}{h} = 0$:

$$\lim_{h\to 0} \frac{(\cos h-1)}{h} = \lim_{h\to 0} \frac{(\cos h-1)}{h} . \frac{\cos h+1}{\cos h+1}$$

$$= \lim_{h\to 0} \frac{(\cos^2 h-1)}{h(\cos h+1)}$$

$$= \lim_{h\to 0} \frac{-\sin^2 h}{h(\cos h+1)}$$

$$= \lim_{h\to 0} \frac{\sin h}{h} . \lim_{h\to 0} \sin h . \lim_{h\to 0} \frac{-1}{(\cos h+1)} = 1 \times 0 \times (-\tfrac{1}{2}) = 0$$

5. **Other trig derivatives**

$$\frac{d}{dx}(\tan x) = \frac{d}{dx}\left(\frac{\sin x}{\cos x}\right)$$

$$= \frac{\cos x\cos x+\sin x\sin x}{(\cos x)^2}$$

$$= \frac{1}{\cos^2 x}$$

$$= \sec^2 x$$

$$\frac{d}{dx}(\operatorname{cosec} x)$$

$$= \frac{d}{dx}\frac{1}{\sin x}$$

$$= \frac{-\cos x}{\sin^2 x}$$

$$= -\frac{\cos x}{\sin x} . \frac{1}{\sin x}$$

$$= -\operatorname{cosec} x . \cot x$$

Derivatives of $\cot x$ and $\sec x$ follow similarly.